Biases From Spectral Leakage in Remote Sensing of Near-Surface Currents

Stefan Weichert, Benjamin K. Smeltzer, and Simen Å. Ellingsen

***Abstract*— Remotely measuring subsurface water currents from imagery of the wave field has become a much-used technique. We study the biases and errors in such measurements due to spectral leakage and suggest mitigating procedures. Deviations between peak values in the 3-D wave spectrum and the known dispersion relation in quiescent water are extracted and interpreted as current-induced Doppler shifts, from which the subsurface current is inferred. The use of discrete Fourier transforms (DFTs), however, introduces spectral leakage between nearby frequency bins. Analyzing synthetically generated wave data adhering to realistic input spectra, we show that although no current is, in fact, present, spurious currents can be "measured," which can amount to a significant fraction of the phase speed at the spectral peak. We analyze the effects of data tapering, method of Doppler shift extraction, limited wavenumber and frequency resolution, peakedness and angular width of the input spectrum, and average misalignment between waves and Doppler shift velocity (DSV) direction. The narrower the input wave spectrum in frequency and/or direction, the greater the biases become. The use of a window function reduces the severity in nearly all cases, yet mitigates the effects of limited resolution more effectively in space than in time. When a current is present, the absolute biases remain essentially unchanged, when waves and currents are roughly aligned; whereas in the case of a cross current, biases remain significant even for tapered data.**

***Index Terms*— Dispersion, remote sensing, sea measurements, sea surface, signal processing algorithms, spectral analysis, surface waves.**

I. Introduction

THE prospect of measuring currents near the sea surface remotely from above is a highly attractive one. Measuring depth-varying currents in situ by penetrating the surface requires the use of, e.g., buoys, ships, gliders, or fixed instruments, all relatively expensive and able to measure a single point or trajectory at a time, and often struggle to capture currents in the top few meters. In comparison, remote measurement from above can be performed with inexpensive equipment mounted on airborne platforms able to cover larger areas in a short time (see [1]).

By far, the most common source of wave data for this purpose has been measurements using HF or X-band radar [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], primarily mounted on ships. Only observation of the wave phase variation in time and space is required, however, meaning other methods are equally applicable; the use of infrared [17] and polarimetric [18], [19] imaging has been demonstrated, as has regular optical measurements (video) with cameras mounted on quadcopter drones [20] and aircraft [21], [22], [23], [24].

A current varying with depth will affect the wave phase velocity differently for different wavelengths, resulting in a measurable effective Doppler shift, which depends on wavenumber [2] and appears as a shift in the waves' spectral dispersion curve from that observed in quiescent water. Ever more advanced methods have been developed in recent years for inferring the depth profile of the subsurface currents from such measured Doppler shifts [12], [25], [26], yet the task of obtaining these shifts from an observation of the spatiotemporal wave field, by reconstructing the current-modified dispersion surface in the frequency–wavenumber spectrum, is itself a nontrivial task. Typically, the spectrum is divided into wavenumber magnitude bins, whereby the Doppler shifts are found for each bin separately. Perhaps, the most frequently employed are least squares (LS)-based methods (e.g., [4], [8], [27]), whereas the alternative normalized scalar product (NSP) method is also in regular use (e.g., [10], [13], [28]). We compare these methods herein finding NSP to be unequivocally favorable. A further method which we shall not consider here is the so-called polar current shell method (e.g., [13]), recently adapted for this purpose [26], which has similar performance as NSP, but favorable in some circumstances.

One should note that the same questions we seek to answer here, also apply to bathymetry retrieval from the measured wave spectra (see [29] and references therein). The extraction of the water depth from the measured spectrum requires spectral intensity at low wavenumbers, which can be strongly influenced by spectral leakage, as we will show. Since a finite water depth enters the dispersion relation in a direction-independent and multiplicative way in contrast to the additive and anisotropic term due to a current, it is not obvious how the issues discussed herein affect the bathymetry retrieval quantitatively, a question of potential importance which requires further investigation.

A. Outline

In this work, we consider the effects of a strong spectral energy peak and the corresponding effect it has on the extraction of Doppler shifts for wavenumbers in the vicinity

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Stefan Weichert and Simen Å. Ellingsen are with the Department of Energy and Process Engineering, Norwegian University of Science and Technology, 7491 Trondheim, Norway (e-mail: stefan.weichert@ntnu.no; simen.a.ellingsen@ntnu.no).

Benjamin K. Smeltzer is with the Marinteknisk senter, SINTEF Ocean, 7052 Trondheim, Norway.

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of the peak. The energy peak results in spectral leakage to adjacent wave vectors in the wave spectrum and may cause errors in the extracted Doppler shifts—as, perhaps, the clearest example we show that in realistic sea states, spectral leakage can cause a significant spurious current to be "measured" when none is, in fact, present.

The phenomenon of spectral leakage is briefly reviewed in Section II-B, and we discuss how it can be mitigated by data windowing. We then go on to describe a numerical experiment where mock wave data are generated and analyzed with the methods in standard use for remote sensing of currents from wave dispersion, in Section III. Results are reported studying how the spurious current "measurements" depend on the method of Doppler shift extraction (Section III-C), the angular width and peakedness of a directional wave spectrum of the Joint North Sea Wave Project type (JONSWAP) [30] (Sections VI-B and VI-C), the spatial and temporal resolution (Section VI-D), and how the situation changes when a uniform background current is present, in Section VI-E. We finally summarize and give a brief overview of ways, whereby the detrimental effect of spectral leakage can be reduced in practical applications, in Section VII.

II. Background

When remote sensing of subsurface currents from wave dispersion is performed, the input spectrum is an observation of the motion of the water surface resolved in time and space. The method is based on linear wave theory, so only the phase of the waves is required, not the amplitude. The data are typically a monotonic function of the sea surface elevation or its derivative, as a function of position (x, y) and time t. A 3-D Fourier transform is then applied in space and time to obtain a spectral signal as a function of wave numbers $\boldsymbol{k} = (k_x, k_y)$ and frequency ω. We assume that the spatial area and duration of the observation are $L \times L$ and T, respectively, so that the resolution in wavenumber and frequency are, respectively, $\delta k = 2\pi/L$ and $\delta\omega = 2\pi/T$; we will refer to these as a pixel or bin in wavenumber and frequency, respectively. (Note that after we introduce nondimensional units in Section III-A, δk and $\delta\omega$ take the forms $1/L$ and $1/T$, respectively.) We will assume infinitely deep water for simplicity herein.

A. Theory

In a wave spectrum, the spectral signal is concentrated near the dispersion relation $\omega = \omega_{\mathrm{DR}}(\boldsymbol{k})$. The methods for sensing the subsurface current now extract a measured function $\omega(\boldsymbol{k})$. If a current with moderately strong depth dependence $\boldsymbol{U}(z) = (U(z), V(z))$ is present, the dispersion relation for a wave with wave vector $\boldsymbol{k}$ is well approximated as $\omega = \omega_{\mathrm{DR}}(\boldsymbol{k}; \tilde{\boldsymbol{c}})$ with the dispersion function

$$\omega_{\mathrm{DR}}\left(\boldsymbol{k}; \tilde{\boldsymbol{c}}\right) = \omega_0(k) + \boldsymbol{k} \cdot \tilde{\boldsymbol{c}} \tag{1}$$

where $k = |\boldsymbol{k}|$, $\tilde{\boldsymbol{c}}$ is the Doppler shift velocity (DSV) due to the presence of a subsurface current and the dispersion relation in deep, and quiescent water is

$$\omega_0(k) = \sqrt{gk}. \tag{2}$$

Inversion methods to infer depth-dependent velocity profiles from the measured spectrum are based on the approximation [2], [31]

$$\tilde{\boldsymbol{c}} = 2k \int_{-\infty}^{0} \boldsymbol{U}(z) e^{2kz} \mathrm{d}z. \tag{3}$$

If a current is present, which is uniform in depth, the resulting Doppler shift will be independent of $\boldsymbol{k}$, while, conversely, a Doppler shift, which varies with $\boldsymbol{k}$, implies the presence of a current, which varies as a function of z. We shall see that biases in the measured Doppler shift $\tilde{\boldsymbol{c}}$ due to spectral leakage typically vary significantly with $\boldsymbol{k}$, and hence, the spurious currents, which are "measured," will have a nontrivial depth dependence. We do not pursue this question in detail.

The presence of the current, thus, introduces an observable Doppler shift corresponding to the addition of a phase speed $\tilde{\boldsymbol{c}}(k)$ to the phase velocity. Remote sensing of the depth-varying current $\boldsymbol{U}(z)$ is then possible by measuring $\tilde{\boldsymbol{c}}$ and inverting (3) using one of several methods available as reviewed in [1]. Methods for extracting $\tilde{\boldsymbol{c}}$ from a measured spectrum are reviewed and compared in Section III-C.

Following Smeltzer et al. [26], we define the instructive quantities $\delta c_{\delta\omega}$ and $\delta c_{\delta k}$ as follows:

$$\delta c_{\delta\omega} = \frac{\delta\omega}{k}; \qquad \delta c_{\delta k} = \frac{\partial\omega}{\partial k}\frac{\delta k}{k} \tag{4}$$

which estimate the change in the predicted phase velocity c due to moving the dispersion surface (1) by δk along the wavenumber axis in the spectrum, or by $\delta\omega$ along the frequency axis. These are, thus, approximate measures of the uncertainty in velocity measurement introduced by limited wavenumber and frequency resolution, respectively. Further discussion may be found in [26, Sec. 4.2.1].

B. Spectral Leakage and Windowing

Assume that a continuous signal (in space or time) $\tilde{P}_0(t)$, the "true" signal, is measured during a finite period of duration T. A sharp cutoff at the beginning and end of the measurement is equivalent to multiplying $\tilde{P}_0$ by a discontinuous top-hat function $w_{\mathrm{box}}(t)$, which is 1 within a time interval of length T and zero outside this "window." Multiplication with such a window is equivalent to a convolution of the spectrum $P_0 = \mathcal{F}\{\tilde{P}_0\}$ with a sinc function in frequency space (e.g., [32])

$$P_{\mathrm{measured}}(f) = P_0(f) * T\,\mathrm{sinc}(fT) \tag{5}$$

where $\mathrm{sinc}(a) = \sin(\pi a)/\pi a$, which is the Fourier transform of the top-hat function. The result is a blurring of the spectrum: "true" spectral components will appear as spectral intensity not only in the frequency bins closest to the actual frequency of said component, but also on neighboring ones that are more than one bin away, with an intensity decreasing with distance. One can alleviate this by replacing the "box" window with a window of choice, simply by multiplying the acquired data $\tilde{P}(t) = \tilde{P}_0 w_{\mathrm{box}}(t)$ with said window function $w(t)$. If the chosen $w(t)$ also vanishes outside the measurement interval, the window replaces the top hat. The choice of the optimal window function much depends on the situation and data at hand.

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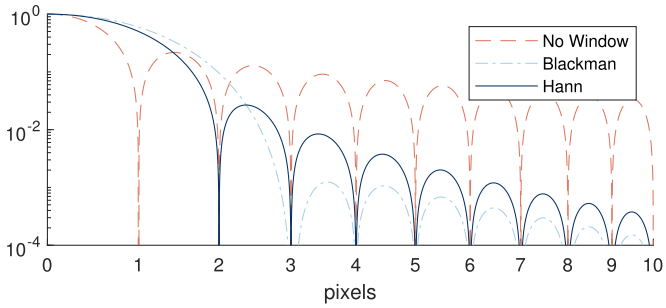


Fig. 1. Fourier transform $\mathcal{F}\{w\}(f)$ of a selection of common window functions $w(t)$. A frequency pixel is $\delta\omega = 2\pi/T$, normalized so that $\mathcal{F}\{w\}(0) = 1$. This illustrates how long-range spectral leakage can be mitigated by tapering the measured signal. The drawback of increased short-range leakage (1–2 pixels) is often negligible.

The Fourier transforms $\mathcal{F}\{w\}(f)$ of some common window functions are shown in Fig. 1 (see [33] for a more comprehensive comparison). When taking a discrete Fourier transform (DFT) of $w(t)$, every real frequency component f will generate spectral intensity in frequency bins of width δf centered at $n\delta f$, $n \in Z$, with intensity $\mathcal{F}\{w\}(n\delta f - f)$. For example, for the *Blackman window* shown in Fig. 1, a frequency coinciding with a DFT frequency, i.e., $f = m\delta f$, one obtains nonzero spectral intensity only for $n - m = 0$–2, leading to no long-range leakage whatsoever. Only the central lobe is sampled in frequency space. For any real spectrum, most frequencies will not coincide with a DFT frequency, and so, the side lobes give rise to long-range leakage. Now, the central lobe of the *no window* (or top hat) case is only one pixel wide, but the leakage is very long range, i.e., spectral intensity decays slowly and stays well above 1% for more than 20 pixels. In contrast to this, the *Blackman* window decays rapidly to 10^{-3}, at the cost of increasing spectral leakage into the two nearest bins, 1 and 2. The best choice of the windowing function is usually a compromise between suppressing long-range leakage and blurring the spectrum (short-range leakage). In this work, a Hann window, defined as

$$w_H(t) = 0.5 - 0.5\,\cos(2\pi\, t/T) \tag{6}$$

is used, as it suppresses the long-range spectral leakage to less than 1% and leaks significantly into frequencies less than two pixels away. In Appendix A, an example of extracted DSVs with different windowing functions is given to illustrate how the choice affects the results.

The data in this work are surface elevation $\zeta(x, y, t)$ measured in three dimensions x and y and time t, which we premultiply by a 3-D Hann window constructed as follows: $w(x, y, t) = w_H(x)w_H(y)w_H(t)$ prior to subjecting it to a discrete 3-D fast Fourier transform (3DFFT) (it is understood that L replaces T in (6) when the argument is x or y). The new signal $\zeta(x, y, t)w(x, y, t)$ goes smoothly to zero at the edges of the domain of observation, in our case, the square area with sides L and time duration T.

III. Methods

We proceed by producing synthetic surface elevation data $\zeta(x, y, t)$ by superposing random linear plane waves of wavenumber k and direction θ from chosen spectra with varying properties, as detailed in Section III-B. Each wave is given a uniformly distributed random initial phase, and the frequency ω is found from (1), whereupon the waves are propagated in time.

These wave "observations" are, of course, idealized, since different methods for obtaining the actual surface elevation $\zeta(x, y, t)$ or true spectrum from field data each come with their individual challenges and limitations, an ongoing field of research in its own right. Taking such practical challenges into account is beyond the scope of this work, and we use the ideal data to isolate the effects of spectra leakage in the data analysis.

We mostly consider the case of quiescent water, i.e., there is no background current, and any Doppler shifts "measured" from the spectrum are spurious and purely a consequence of spectral leakage. We also consider the case where a constant background current $\boldsymbol{U}_0$ is present, in Section VI-E. Biases now manifest as deviations of the observed DSVs from the correct value.

A. Dimensionless Quantities

Dimensional quantities will be denoted by a superscript asterisk, and all other quantities are nondimensionalized. The reference length scale and time scale are defined based on a characteristic wavenumber k_0^* and its corresponding angular frequency in quiescent water $\omega_0^* = (gk_0^*)^{1/2}$. Thus, e.g., $T = T^*\omega_0^*/2\pi$, $L = L^*k_0^*/2\pi$, $k = k^*/k_0^*$, $\omega = \omega^*/\omega_0^*$, and $U = U^*(k_0^*/g)^{1/2}$. Unless specified otherwise, $\omega_0^*(k_0^*)$ is taken to be the location of the peak of the energy spectrum, i.e., the spectral peak is at $\omega = 1$ by definition; hence, the peak in wavenumber space is close to $k = 1$.

B. Wave Spectrum

We generate wave fields from commonly used realistic model spectra with varying directional broadness, assuming the form

$$\hat{S}(\omega, \theta) = S(\omega)\, f(\theta) \tag{7}$$

where θ is the angle between $\boldsymbol{k}$ and the x-axis, $\cos\theta = \boldsymbol{k}\cdot\boldsymbol{e}_x/k$. We use the JONSWAP spectrum [30]

$$S(\omega) = \tilde{N}\omega^{-5}\exp\left[-\frac{5}{4}\omega^{-4}\right]\gamma^{r(\omega)} \tag{8}$$

with

$$r(\omega) = \exp\left[-\frac{1}{2}\left(\frac{\omega - 1}{\sigma}\right)^2\right]. \tag{9}$$

For our purposes, the value of $\tilde{N}$ is not of importance, and we set $\tilde{N} = 1$. The parameter σ is

$$\sigma = \begin{cases} 0.07, & \text{if } \omega \le \omega_p \\ 0.09, & \text{if } \omega > \omega_p \end{cases} \tag{10}$$

and the peakedness parameter γ is varied (see Section VI-C). The energy spectrum for a selection of peakedness values γ is depicted in Fig. 2. The angular distribution is taken as a

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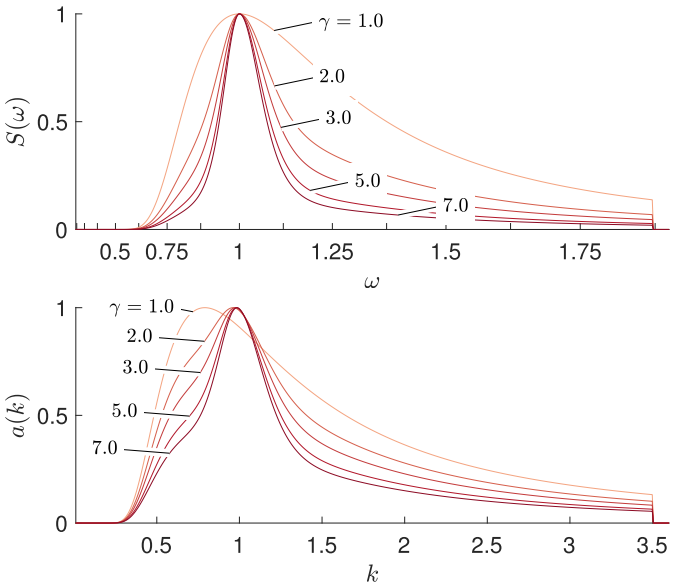


Fig. 2. Top: energy spectrum $S(\omega)$ (8) for a few values of the peakedness parameter γ. The ω-axis is scaled quadratically to match the range of the bottom graph, as $k = \omega^2$. Bottom: amplitudes $a(k, \theta = \theta_0)$ (12) of superposed wave components. The spectra are normalized with respect to their peak value.

cosine square with a full width (distance of first roots) of $\Delta\theta$, i.e.,

$$f(\theta) = \begin{cases} \cos^2\left(\pi \dfrac{\theta - \theta_0}{\Delta\theta}\right), & |\theta - \theta_0| \le \Delta\theta/2 \\ 0, & \text{else.} \end{cases} \tag{11}$$

We use this spectrum to prescribe the amplitudes $a(\boldsymbol{k})$ of the superposed wave components in accordance with [34] as

$$a(\boldsymbol{k}) = \sqrt{2\, k^{-3/2}\, \hat{S}(\omega(k), \theta)\delta k} \tag{12}$$

on an evenly spaced grid in k_x–k_y, with spacing $\delta k_x = \delta k_y = \delta k$. In addition, we set $a(\boldsymbol{k}) = 0$ for $k/k_p > 3.5$.

C. Normalized Scalar Product

Two methods are in common use for extracting Doppler shifts from the measured wave spectrum, LS methods (e.g., [4], [8], [27]) and the NSP method (e.g., [10], [13], [28]). While there exist a wide range of extensions and sophistications to LS methods, we focus our attention on the NSP method. However, since two of the most famous works on spectrum-based current extraction [4], [8] use a simple form of the LS method, we also give a brief comparison of their performance in Appendix C.

We employ the DSV extraction method as implemented by Smeltzer et al. [26]; see [10], [13], and [28] for details on the NSP method more generally.

The starting point in either method is a measured, spatiotemporally resolved free surface $\eta(x, y, t)$ and its power spectrum obtained via a DFT, $P(\boldsymbol{k}, \omega) = |\text{FFT}[\eta(x, y, t)]|^2$. For each wavenumber k_i in a list, the spectral intensity on a cylindrical surface with radius k_i centered around the ω-axis is defined as follows:

$$F_i(\boldsymbol{k}, \omega) = \sqrt{P(k_i \cos\theta, k_i \sin\theta, \omega)} \tag{13}$$

where θ is the azimuth angle in the k_x–k_y plane. The implementation of the algorithms used is formulated in Cartesian coordinates; however, it is illustrative to use cylindrical coordinates for the following conceptual considerations. More details of the implementation are given in Appendix B.

To find the effective DSV $\tilde{\boldsymbol{c}}_i = \tilde{\boldsymbol{c}}(k_i)$, first, a characteristic function G is defined that contains the components of $\tilde{\boldsymbol{c}}_i$ as free parameters

$$G_i\left(\theta, \omega; \tilde{\boldsymbol{c}}_i\right) = G_i^+\left(\theta, \omega; \tilde{\boldsymbol{c}}_i\right) + G_i^-\left(\theta, \omega; \tilde{\boldsymbol{c}}_i\right) \tag{14}$$

where

$$G_i^{\pm}\left(\theta, \omega; \tilde{\boldsymbol{c}}_i\right) = \exp\left[-2\left(\frac{\omega \pm \omega_{\text{DR}}\left(\theta; k_i, \tilde{\boldsymbol{c}}_i\right)}{a}\right)^2\right]. \tag{15}$$

The NSP N_i of the vectors F_i and G_i is now maximized for each value of i by varying the two components of $\tilde{\boldsymbol{c}}_i$; it is calculated as follows:

$$N_i\left(\tilde{\boldsymbol{c}}\right) = \frac{\langle G_i F_i \rangle}{\langle G_i \rangle \langle F_i \rangle} \tag{16}$$

where $\langle \ldots \rangle$ refers to an integral over all θ and ω. In other words, G_i can be thought of as a cosine in θ with offset $\omega_0(k_i)$, amplitude $k_i \tilde{c}$, and phase shift defined by the direction of $\tilde{\boldsymbol{c}}$; its overlap with the measured intensity on the cylinder surface is maximized to find the best DSV $\tilde{\boldsymbol{c}}(k_i)$. This optimization step is performed using the Nelder–Mead simplex method [35].

IV. NSP Method and Spectral Leakage

For multidimensional data, spectral leakage is most prominent in the directions parallel to the coordinate axes. Consider, for example, a 2-D signal $\eta(x, y)$ on a rectangular domain and its Fourier transform $\tilde{\eta}(k_x, k_y) = \mathcal{F}\{\eta\}$. Since both x and y are within a finite range, the effective window $w(x, y)$ is a product of top-hat windows in the x- and y-directions, respectively, i.e., $w = w_1(x)w_2(y)$. The Fourier transform of such a product is the product of their respective Fourier transforms $\tilde{w}(k_x, k_y) = \mathcal{F}\{w_1(x)\}\mathcal{F}\{w_2(y)\} = \tilde{w}_1(k_x)\tilde{w}_2(k_y)$. Now, since the leakage for a top-hat window falls off as $1/k$, the product $\tilde{w}(k_x, k_y)$ is smallest for a given k when $|k_x| = |k_y|$ and largest when $k_x = 0$ or $k_y = 0$. An illustration of this can be seen in Fig. 3. The extension to three or more dimensions is straightforward.

This (mostly) axes-parallel leakage is helpful in understanding how leakage affects the DSV extraction using NSP.

A fundamental step in the NSP method is to pick out measured spectral intensity on a cylinder surface with radius k_i. One can think of the effect on the NSP as the spectral intensity projecting itself in the principal directions onto the cylinder, with decreasing intensity the further the surface is from the originating spectral intensity. Now, as the NSP method essentially fits a function of the form $\omega_0(k_i) + A_f \cos(\theta - \theta_f)$, with free parameters A_f and θ_f, to the spectral intensity on the cylinder surface defined by k_i, as illustrated in Fig. 4, spurious intensity at $\omega \neq \omega_0(k_i)$ leads to nonzero values for $A_f = \boldsymbol{k} \cdot \tilde{\boldsymbol{c}}$, implying a background current even in the absence of one. In this work, we consider spectra with a single peak at a given wave vector $\boldsymbol{k} = (k_x, k_y)$.

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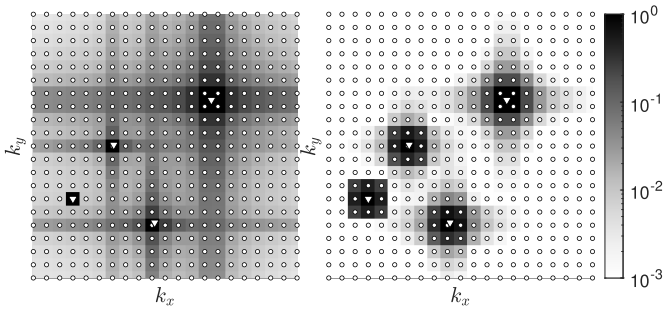


Fig. 3. Illustration of spectral leakage in two dimensions. Left: logarithmic 2-D FFT spectrum of a superposition of four sine waves (i.e., the spectrum is the sum of four delta functions in the $\boldsymbol{k}$ plane), with frequencies denoted by triangles. The circles denote the DFT frequencies, i.e., frequencies natural to the domain. From left to right, the leftmost spectral peak coincides with a DFT frequency, while the rightmost peak lies in the middle of DFT frequencies. Right: same as left, bit with a 2-D Hann window applied before doing the FFT. Note how the Hann window removes the background and increases consistency between spectral peaks, while broadening by roughly one bin in all directions.

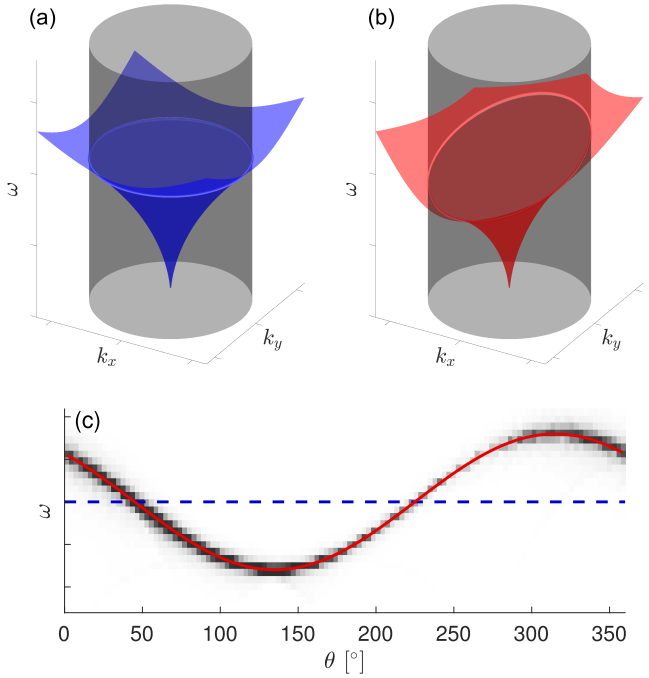


Fig. 4. Illustration of the dispersion relation $\omega = \omega_{\mathrm{DR}}(\boldsymbol{k}; \tilde{\boldsymbol{c}})$ in connection within the nonlinear scalar product method. The dispersion relation (a) in the absence of a current, $\tilde{\boldsymbol{c}} = 0$ and (b) with a current, $\tilde{\boldsymbol{c}} = \boldsymbol{U}_0$. The intersections with a cylinder with radius k_i are highlighted. (c) Spectral intensity is indicated in gray scale, visible near the dispersion curve on the cylinder surface after unrolling it, together with the lines of intersection from the top figures: blue dashed line corresponds to quiescent water; red solid line is the best fit from the NSP method resulting in a nonvanishing DSV.

We now consider the effect of spectral leakage for two distinct cases as follows: 1) with the mean propagation direction θ_0 being parallel to the k_y-axis ($\theta_0 = 90°$) and 2) with θ_0 between 0° and 90°.

A. Leakage for $\theta_0 = 90°$

Consider the extraction of the DSV for a wavenumber $k < 1$, i.e., below the peak of the spectrum. The spectral leakage for the cylinder with radius k (see Fig. 4) will be dominated by short-range spectral leakage of wavenumbers close to k and long-range leakage from the spectral peak. The short-range spectral leakage for the most part will merely blur the spectral intensity on the cylinder, while the long-range leakage from the spectral peak leads to a new, possibly dominating, spectral intensity spot on the cylinder, at the same angle, but a higher frequency. In other words, right above the real spectral intensity. Depending on the ratio of intensities, distance, and width of the characteristic function G, the NSP algorithm may find the correct frequency, the frequency of the projected peak, or a value in between these two to give the best fit. Therefore, the returned DSV will be parallel to the spectrum propagation direction θ_0. Similarly, if $k > 1$ is considered, the projection from the spectral peak will be below the real spectral intensity, giving DSV antiparallel to the spectrum propagation direction θ_0.

B. Leakage for $0° < \theta_0 < 90°$

For $k < 1$, it is now possible for spectral leakage from the peak to have no axes-parallel projection onto the cylinder surface, thus not affecting the extracted DSV for this wavenumber. Short-range spectral leakage and leakage from lower wavenumbers now dominate, making the prediction of DSV direction difficult. For $k > 1$, one can still get an intuition for the effect of leakage. As the peak of the spectrum lies in the first quadrant of the k_x–k_y plane, the shortest distance to a cylinder with radius k is to the quarter of the cylinder also lying in the first quadrant, and we can focus our consideration on that. The leakage is now being projected onto the cylinder at an angle relative to the mean propagation direction θ_0. Therefore, as the real spectral intensity now dominates for $\theta = \theta_0$, the frequency for wavenumbers along θ_0 is found correctly. The perpendicular component of the DSV, however, is determined mostly by the position and relative magnitude of the spurious intensity appearing at angles $\theta \neq \theta_0$. In quiescent water, for example, one would obtain DSVs with directions $\theta \pm 90°$.

Note that these considerations hold true for very narrow spectra, but are to be understood as tendencies for spectra with considerable spectral width $\Delta\theta$, because, for any angle θ, real spectral intensity will mostly dominate over spectral leakage, if present.

V. Parameter Choice and Overview

For an overview and easier referencing, Table I contains the list of test cases presented in this article. Apart from the parameters stated therein, the resolution of the input spectrum $\delta_{\mathrm{in}}k$ and spatial resolution $\delta x = \delta y$ as well as temporal resolution δt had to be set. The spatial resolution was chosen, such that waves of the highest wavenumbers, $k_{\max} = 4$, would be well resolved, $\delta x = 1/28$. Similarly, time resolution was set to $\delta t = 1/14$. The resolution of the input spectrum was set to $\delta_{\mathrm{in}}k \approx 0.341/L$, with L being the domain size. This was deemed small enough to mimic a continuous spectrum.

The domain size and video duration were chosen to be similar to typical parameters found in airborne measurements. For instance, videos used in [36] typically have a field of view of $L^* = 128$–512 m, durations of $T^* = 20$–40 s, and wavenumbers with usable spectral intensity from 0.2 to 2 rad/m (see [36, Fig. 1]). Taking a reference

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wavenumber of $k_0^* = 0.4\,\mathrm{rad/m}$, this gives a parameter range of $T \approx 5$–10 and $L \approx 8$–32.

Equation (14) contains one free parameter, a, that determines the width of the characteristic function. We set $a = 4\,\delta\omega$, with $\delta\omega = 1/T$ being the frequency resolution (see Appendix B for details).

The results shown in Figs. 5–9 display the statistics of the extracted DSVs in terms of the average $\langle|\tilde{\boldsymbol{c}}|\rangle$ and the corresponding standard deviation $\sigma_{\tilde{c}}$ ($\times 2$ for illustrative purposes), calculated from 100 realizations. These quantities are useful, as they represent a mean bias and fluctuation that one has to expect from a single measurement. In addition, where possible, the implied velocity resolutions $\delta c_{\delta\omega}$ and $\delta c_{\delta k}$ defined in (4) are shown for reference.

VI. Results

In this section, we consider the effects of wave-spectral properties, resolution, and data tapering on the spurious Doppler shifts "measured" when no current is present, as well as the effect of a uniform current being present. An overview of the parameter combinations used to obtain the following results is given in Section V.

A. Illustration of the Effect of Windowing (Hann Window)

The mitigating effect of windowing/tapering is well illustrated when considering the influence of the angular spread $\Delta\theta$ of the wavenumber spectrum; simulations with variation both in $\Delta\theta$ and the mean propagation direction θ_0 were performed (see Table I, case θ_0–$\Delta\theta$, for all parameters). A selection of the results is shown in Fig. 5 (top row). A more detailed discussion of the influence of $\Delta\theta$ and θ_0 is given in Section VI-A.

The propagation direction θ_0 shows a strong influence on both the mean bias and its variation. This behavior is likely due to spectral leakage, because spectral leakage appears as "streaks" in the spectrum along the spectral axes k_x, k_y, and ω (see illustration in Fig. 3). The second column in Fig. 5 shows the extracted DSVs of tapered surface elevation data. The mean and random biases are strongly suppressed in all cases, and the dependency on θ_0 is significantly reduced, compared with the analysis without a Hann window. Note that $\Delta\theta = 15°$ represents a rare, very narrow spectrum, but the described effects are visible for all spectral widths, decreasing with increasing $\Delta\theta$.

We observe that for smaller wavenumbers ($k < 1$), using a Hann window does not mitigate the effects of spectral leakage as strongly as for $k > 1$. This is because the Hann window suppresses long-range spectral leakage (more than 1–2 pixels), while increasing the short-range spectral leakage (the central lobe of the Hann window has a width of two bins instead of one for no window). The "steepness" $\partial\omega/\partial k$ grows with decreasing k, meaning that spectral intensity that leaks in the k-direction appears as a strong broadening in the ω-direction at neighboring k values. In our simulations, the spectrum has a peak at $\omega = 1$ (thus, close to $k = 1$) and quickly decreases toward $k \approx 0.5$. Therefore, the combined effects of a steep slope in the energy spectrum and the dispersion relation cause the short-range spectral leakage to be most prominent for $k < 1$.

The effect of a steep spectral slope—discussed further in Section VI-C—can be illustrated by considering that at, say, $k = 0.6$, the spectrum shows intensity at $\omega_0(0.6)$, but also intensity leaked from $\omega_0(0.6 + \delta k)$ and $\omega_0(0.6 - \delta k)$, the latter of which is much less than the former. The algorithm, therefore, finds a frequency between $\omega_0(0.6)$ and $\omega_0(0.6+\delta k)$, which gives a nonzero DSV (see also Section IV).

B. Influence of the Mean Propagation Direction and Angular Spread

To study the influence of the angular width on the mean and random bias in the extracted DSVs, simulations were run with three propagation directions $\theta_0 \in \{90°, 75°, 60°\}$ and a range of spectral widths $\Delta\theta$, of which two representative examples ($15°$, $60°$) are shown in Fig. 5 (see Table I, case θ_0–$\Delta\theta$, for all parameters).

All cases show that an increase in spectral width $\Delta\theta$ significantly decreases both mean and random biases across all wavenumbers. Results for $\Delta\theta > 60°$ are not shown here, as no notable further improvements were observed.

The reduction in angular spread $\Delta\theta$ leads to an increase in spurious DSVs $\tilde{\boldsymbol{c}}$, because the wave components of a narrow spectrum are mostly influenced by the current component parallel to the mean propagation direction θ_0. The narrower a spectrum is, the smaller an observable change in frequency due to the perpendicular current component $\tilde{\boldsymbol{c}}_\perp$ becomes. Therefore, the influence of false spectral intensity from any source (e.g., noise, aliasing, higher harmonics, and spectral leakage) outside the real spectrum can lead to strong perpendicular DSV components when the spectrum is very narrow. For a more detailed explanation, see Fig. 5 where we see the combined influence of a noisy spectrum and spectral leakage.

Note that all cases exhibit negligible spurious DSVs around $k \approx 1$. This is simply a consequence of the spectral peak lying near this value.

The dependency of results on the mean propagation direction θ_0 was also investigated in a second manner: the videos were rotated numerically by angles θ_{rot} up to $45°$ before repeating the DSV extraction. This is equivalent to rotating the camera in an experiment. Pairwise comparison of results with the same values of $\theta_0 + \theta_{\mathrm{rot}}$ shows no significant difference.

Two conclusions can be drawn: first, the only effect θ_{rot} can have on physical quantities is a trivial rotation of the coordinate system, so any other θ_{rot} dependencies can be traced back to spectral leakage, which mainly occurs parallel to the video's axes, thus breaking rotational symmetry. Second, we will see in Section VI-E that when a current is present, the measurements can indeed differ after camera rotation, hence rotating the camera or data provides a useful check for spectral leakage.

The use of a Hann window prior to DSV extraction greatly mitigates the effects of spectral leakage, as discussed before, pushing even the narrow spectrum cases down to subresolution (with respect to implied velocity resolutions $\delta c_{\delta\omega}$ and $\delta c_{\delta k}$).

C. Influence of Peakedness γ

Depending on how developed a sea is, the best fit for the frequency spectrum uses a peak enhancement factor γ between 1 and 7, where 3.3 is a commonly used value for

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TABLE I

OVERVIEW OF PARAMETER COMBINATIONS. θ_0: MEAN PROPAGATION DIRECTION OF INPUT WAVENUMBER SPECTRUM, $\Delta\theta$: ANGULAR SPREAD OF INPUT WAVENUMBER SPECTRUM [SEE (11)], γ: PEAKEDNESS PARAMETER [SEE (8)], L: DOMAIN SPATIAL DOMAIN SIZE, T: VIDEO DURATION, AND U: BACKGROUND CURRENT. ALL PARAMETERS ARE NONDIMENSIONAL; SEE SECTION III-A

	θ_0	$\Delta\theta$	γ	L	T	U	figure	section
case-NSP-LS	75°	60°	3.3	10	20	0	11	C
case-θ_0-$\Delta\theta$	60°, 90°	15°...60°	3.3	10	20	0	5	VI-A,VI-B
case-γ	60°	15°...60°	1...7	10	20	0	6	VI-C
case-L-T	90°	60°	3.3	5, 10, 20	10, 20, 80	0	7	VI-D
case-U	90°	60°	3.3	10	20	0...0.15	8, 9	VI-E

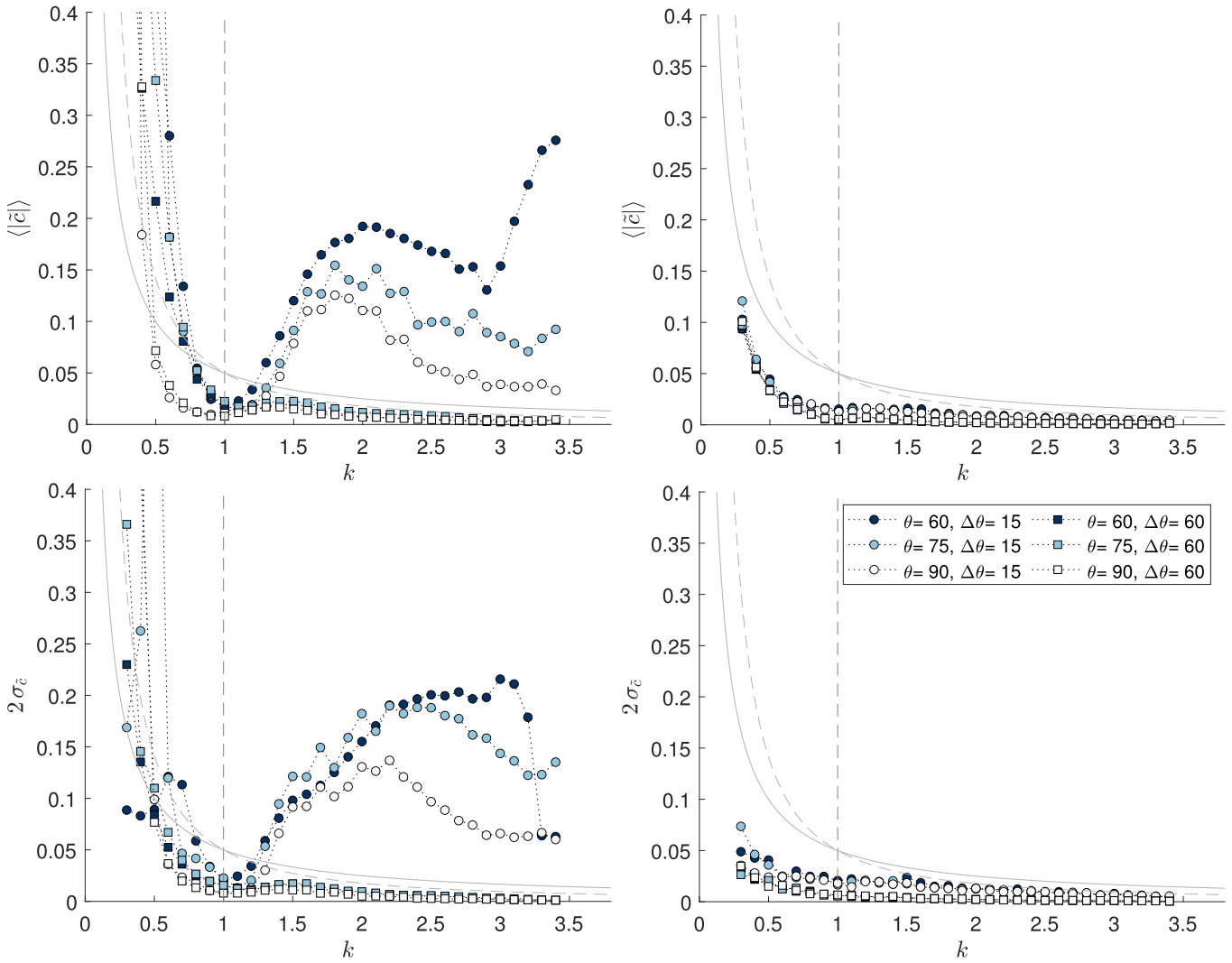


Fig. 5. DSVs $\tilde{\boldsymbol{c}}$ in units of the peak phase velocity for the parameter set of case θ_0–$\Delta\theta$ (see Table I), extracted using the NSP method. The DSVs were extracted for 100 realizations with (right) and without (left) applying a Hann window beforehand. The top and bottom graphs show the biases in DSVs in terms of the average (top) and standard deviation (bottom) of the absolute value $\tilde{c} = |\tilde{\boldsymbol{c}}|$. The solid and dashed lines correspond to the velocity resolutions implied by frequency and wavenumber resolution, respectively.

most applications [37]. Recently, Mazzaretto et al. [38] found that a global mean of $\gamma \approx 2.4$ is better suited.

Since the choice of γ only affects the spectrum in a small range around the peak, it also offers itself as a tool to examine the influence of steep gradients in the spectrum. We, therefore, compare the DSVs for $\gamma \in \{1, \ldots, 7\}$ (see Table I, case γ, for all parameters).

The results presented in Fig. 6 show that for spectra with small angular spread, a more strongly peaked spectrum (higher value for γ) causes increased biases (mean and random) across all wavenumbers except for a small range around the spectral peak at $k \approx 1$, where the relative increase in spectral intensity reduces the influence of spectral leakage from wavenumbers $k \neq 1$. A wider angular spread of the spectrum, on the other hand, mitigates this to a large degree, especially toward higher wavenumbers, where the influence of spectral leakage from the peak is nearly eliminated and subresolution biases are achieved. For angular spreads larger than $\Delta\theta > 45°$, the DSVs become independent of $\Delta\theta$, though this also changes with spectral resolution.

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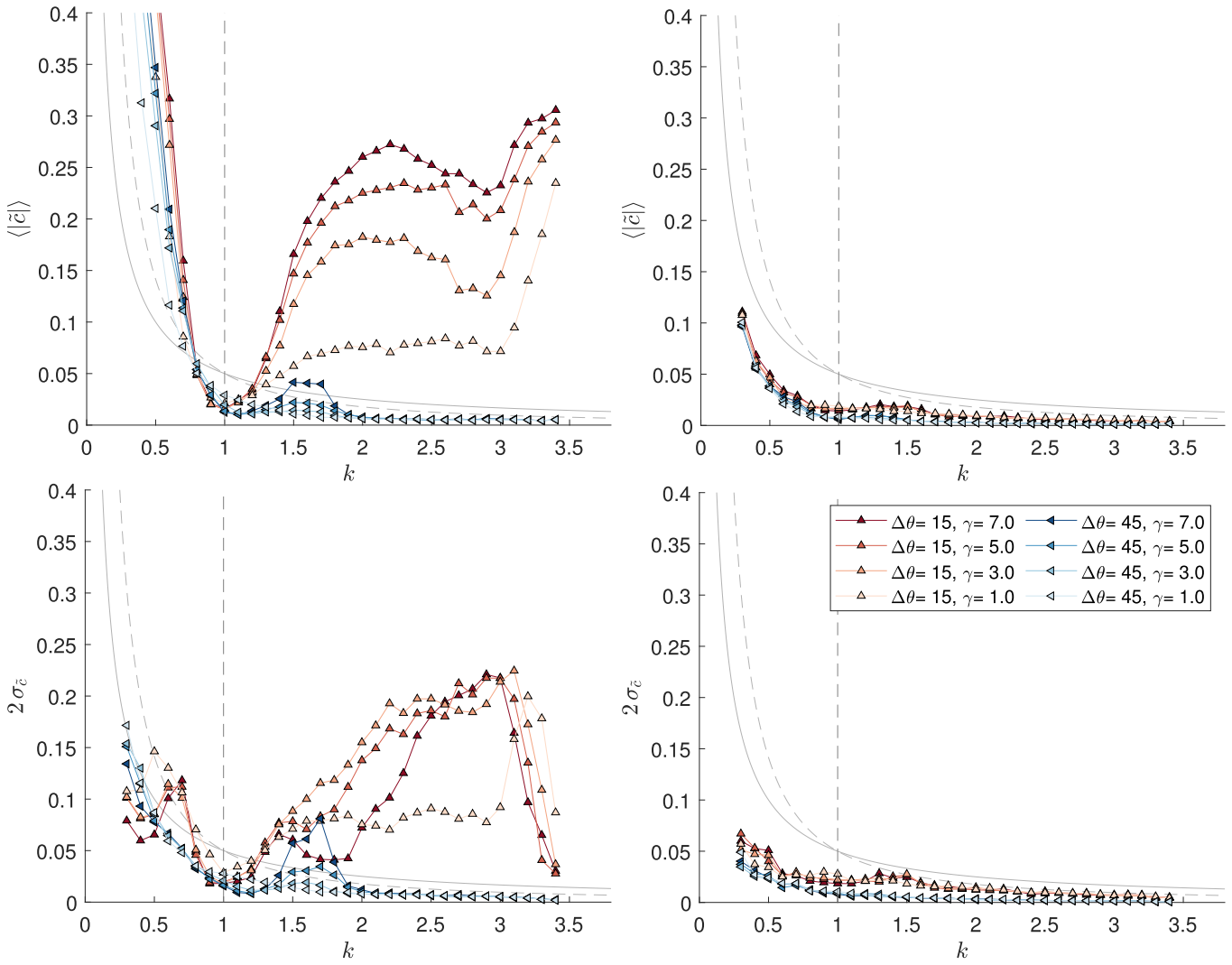


Fig. 6. Same as Fig. 5 but for a selection of parameter combinations from the parameter set of case γ (see Table I).

Toward lower wavenumbers ($k < 1$), without a Hann window, any extracted DSV is unreliable, as spectral leakage dominates the result due to the exponential decrease of "real" spectral intensity toward $k = 0.5$ and the steepness of the dispersion relation (see previous discussion in Section VI-A).

Once again, the use of a Hann window improves the results so far that random and mean biases are subresolution even for strongly peaked ($\gamma = 7$), narrow ($\Delta\theta = 15°$) spectra, effectively eliminating the dependence on $\Delta\theta$ and, to a degree, on resolutions (see Section VI-D).

D. Influence of Resolutions δk, $\delta\omega$

The range of spectral leakage is constant in terms of the number of pixels or bins affected (see Fig. 1), and thus, the leakage range in k (and ω) is determined by the resolution δk (and $\delta\omega$).

We, therefore, vary the spatial size L and temporal duration T of the videos to change the frequency resolution $\delta\omega = 1/T$ and wavenumber resolution $\delta k = 1/L$ to observe their influence on the DSV extraction (see Table I, case L–T, for all parameters).

For a spectrum with an angular spread of $\Delta\theta = 60°$ and a mean propagation direction of $\theta_0 = 90°$, simulations were performed with a range of video lengths $T = 5$–80 and domain sizes $L = 5$–20. The resulting DSVs are shown in Fig. 7 (note that the results for $T < 10$ are not shown, as they did not show relevant differences compared with results for $T = 10$).

Clearly, for untapered data, both improving δk or $\delta\omega$ reduces the biases in DSVs, albeit improving the wavenumber resolution δk has a stronger influence.

Note how the maximum in biases in Fig. 7 moves to higher k with decreasing T. This is due to the characteristic function G having a width a proportional to frequency resolution, $a = 4\delta\omega$. When this width is reduced, the intensity that leaks from the spectral peak onto the cylinder at $k > 1$ can fall outside the reach of the characteristic function, thus reducing or removing its influence.

When the data are tapered using a Hann window, this effect is mostly eliminated, as the long-range spectral leakage is heavily suppressed. Moreover, the influence of frequency resolution $\delta\omega$ is strongly reduced (graphs of the same color group together).

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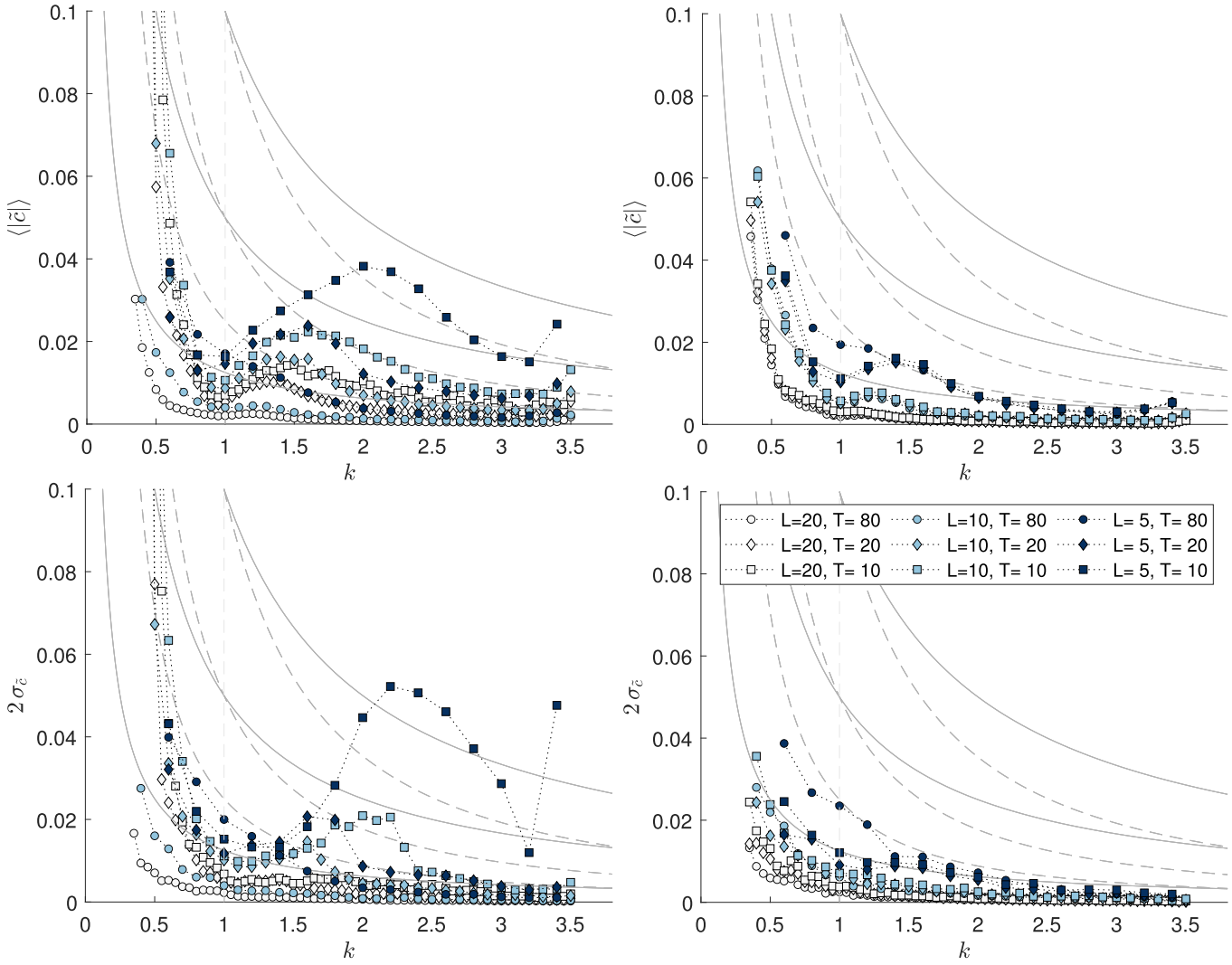


Fig. 7. Same as Fig. 5 but with parameter set, case L–T (see Table I). As in Fig. 5, the solid and dashed lines correspond to the implied velocity resolutions from frequency and wavenumber resolution, respectively. From top to bottom, the resolution increases, i.e., δw or δk becomes smaller.

An exception to the rule of thumb that longer videos are always better can be seen for the longest videos $T = 80$ on the smallest domain $L = 5$. Here, the biases are actually greater than for the shorter cases with $T \leq 40$. This is due to the characteristic function G becoming narrow enough to not encompass the width of spectral intensity in the ω-direction. In one dimension, this would be proportional to $\delta\omega$, but in 2-D or 3-D can be dominated by leakage in k, as this effective broadening in ω scales with $\mathcal{O}(\partial\omega/\partial k\ \delta k) = \mathcal{O}(\delta k/\sqrt{k})$.

Note, however, that the results presented in Fig. 7 all show biases well below the implied velocity resolutions, provided the data are tapered using a Hann window. If one has to choose between increasing the domain size or the video duration, it is clear that an increase in domain size will give the most benefit.

E. Influence of a Background Current U_0

The cases shown above are without a background current. The presence of a spatiotemporally uniform current $\boldsymbol{U} \neq 0$ breaks the rotational symmetry of the dispersion relation because of the additional, angular-dependent term $\boldsymbol{k} \cdot \boldsymbol{U}$ in (1).

To see the effect of a current on the biases in the extracted DSVs, we assume a (vertically and horizontally) constant current $\boldsymbol{U}$ with directions $\theta_U = 90°, 0°$, and $-90°$ and current strengths in the range $|\boldsymbol{U}| = 0.025$–0.15 (see Table I, case U, for all parameters). The input wavenumber spectrum has an angular spread of $\Delta\theta = 60°$, and a peakedness of $\gamma = 3.3$, representing a realistic scenario. The mean propagation direction is held at $\theta_0 = 90°$, resulting in the following, crossing and opposing current, respectively.

The extracted DSVs are shown in Fig. 8. For $k > 1$, we see a bias toward wave-opposing DSVs, while for $k < 1$, we see a bias toward wave-following DSVs. This can be seen from over-/underestimations for $\boldsymbol{U}$ antiparallel/parallel to the wave propagation direction, respectively. The absolute magnitude of errors is similar to the case of quiescent water (see Section VI-D). In the case of a pure cross current, however, this also results in a turning of the DSVs toward $\theta_{\tilde{\boldsymbol{c}}} = \pm 90°$ for wavenumbers around the spectral peak $k \lesssim 1.5$, as can be seen in the third row of panels in Fig. 8. In particular, where the spectral intensity falls off to zero rapidly ($k \approx 0.5$), leakage from the peak of the spectrum can dominate the signal, biasing

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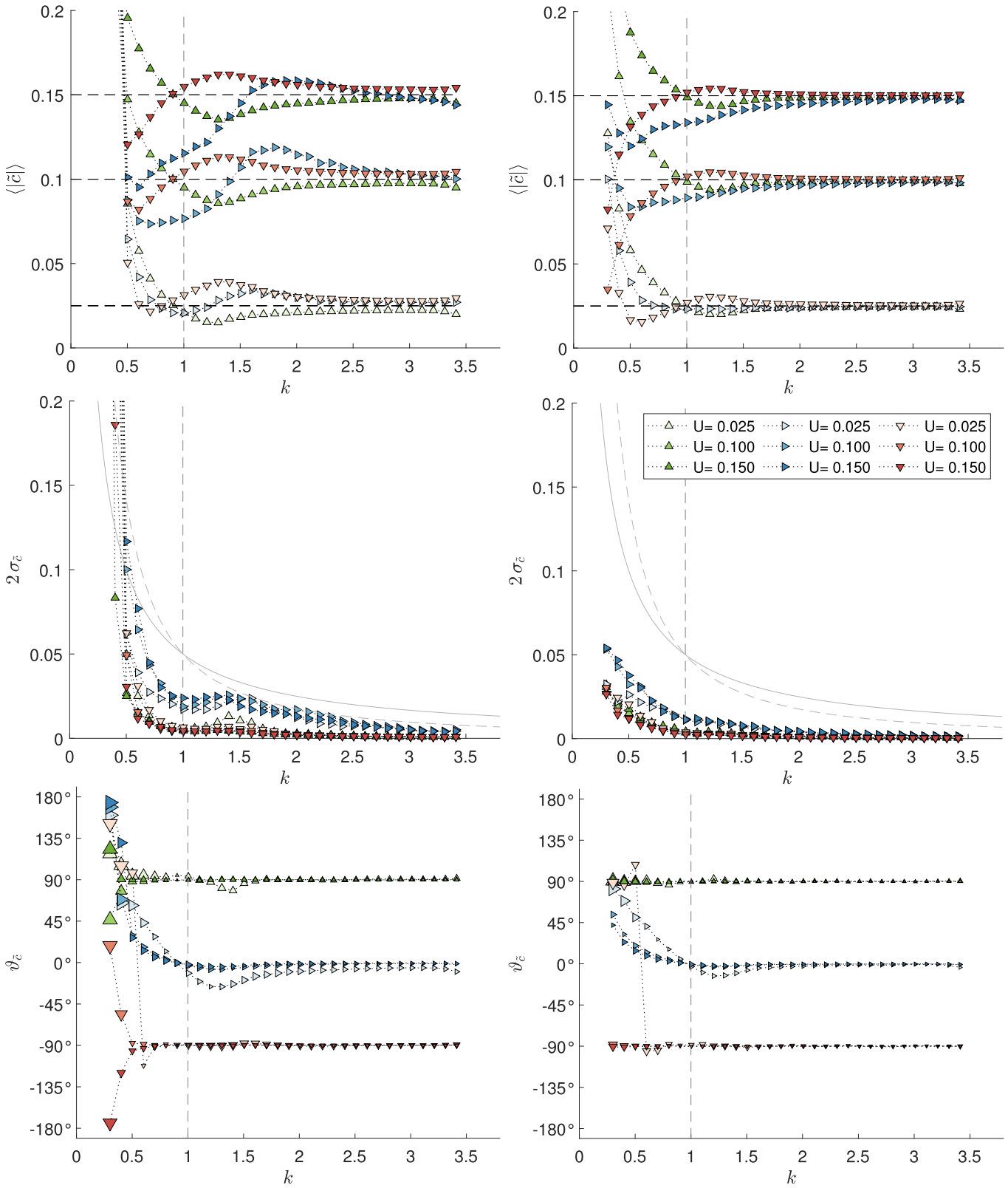


Fig. 8. Same as Fig. 5 but with parameter set, case U (see Table I). The dashed horizontal lines represent the three background velocity values. Deviations from these are spurious. The third row shows the direction $\theta_{\tilde{c}}$ of the average DSV.

the direction of DSVs toward the wave propagation direction, i.e., 90°.

Tapering the data using a Hann window mitigates these biases in amplitude and direction as in the case of quiescent water. For a cross current, however, the mean bias remains significant, i.e., on the order of 10% of the phase velocity at the spectral peak and unusually also in the vicinity of the spectral peak.

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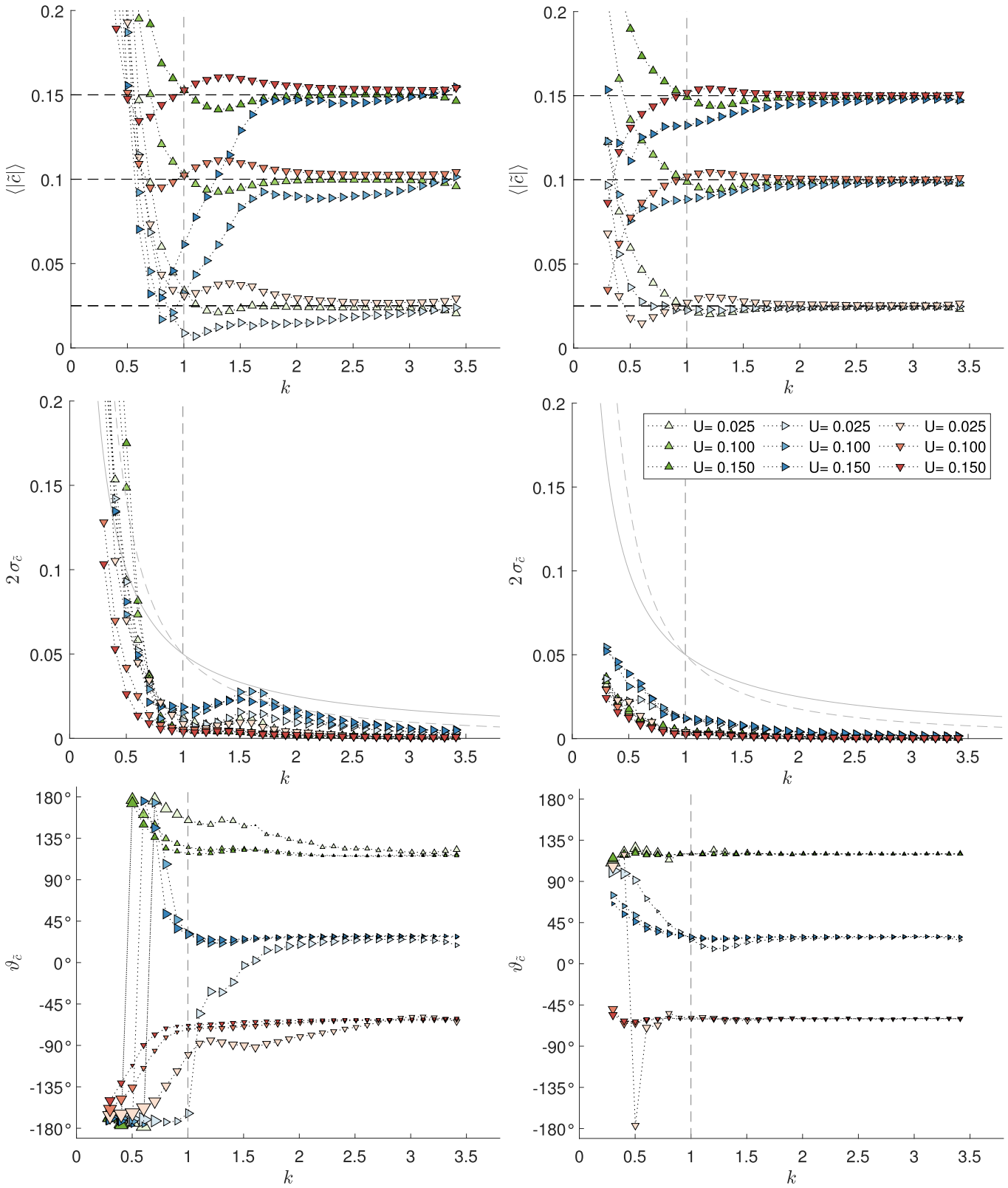


Fig. 9. Same as Fig. 8 but with a 30° rotated video. Note how the case of cross currents (blue right facing triangles) results in unusable DSVs $\tilde{\boldsymbol{c}}$ especially around the spectral peak, when the data are untapered (top left).

The error in the direction $\theta_{\tilde{\boldsymbol{c}}}$ of DSVs for wavenumbers below the peak is also reduced significantly for following and opposing currents. In the case of a cross current, these errors remain significant (10°–45°), if the current is relatively weak or the wavenumber goes toward $k = 0.5$.

As mentioned in Section VI-B, turning the camera in an aerial measurement can reveal the influence of spectral leakage (one could also rotate the resulting images, bearing in mind that this is not, in general, a lossless operation and might also reduce the field of view). Fig. 9 shows the extracted DSVs for

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a case with the same parameters as in Fig. 8, but with current and spectrum rotated by 30° (similar results were obtained for 15°–45°).

For the untapered data, we find even stronger mean biases after rotation, especially for a cross current. Here, the DSVs around the spectral peak at $k \approx 1$ drop below 50 % of the background current, usually an unacceptable level of error. However, the use of a Hann window again mitigates this, to the point where the results of different rotation angles become virtually indistinguishable.

VII. Conclusion

We have investigated biases from spectral leakage in remote sensing of currents from analysis of wave spectra. Apparent, spurious Doppler shifts in the phase velocity are observed even in the absence of a current, and we analyze how these depend on wave spectrum and the properties of the signal processing procedure. Synthetically generated surface elevation data were used to simulate a random sea state adhering to a JONSWAP wave spectrum [30] with a $\cos^2$ directional distribution, resulting in cubes of data (videos). These were subsequently analyzed with different methods in common use to extract the Doppler shift velocities (DSVs). Following [26], an appropriate measure of random errors and biases are the "Doppler shift resolutions" corresponding to the change in inferred velocity due to a shift of one pixel in wavenumber or frequency.

A comparison of the normalized scalar product (NSP) approach (e.g., [26]) and a least squares (LS) method for extracting DSVs showed that the former is preferable in all cases, and was, therefore, used for all subsequent analysis herein.

Assuming the simplest case of quiescent water (i.e., no current), a complex interplay is found between wave-spectrum width and peakedness, and the wave vector and frequency resolution, together affecting the nature and extent of spurious Doppler shift "measurements." Spectral leakage causes greater problems when the wave spectrum is strongly peaked and highly directional, so that areas of the observed frequency–wave vector spectrum, which are important to Doppler shift extraction, have very low signal. Conversely, when the angular spread is wide, $\Delta\theta > 60°$ ($\Delta\theta$: full width of angular distribution), spurious DSVs are small, i.e., subresolution. For strongly directional spectra $\Delta\theta < 60°$, severe biases emerge, with amplitudes on the order of the group velocity, depending on the JONSWAP peakedness parameter γ. The biases are sensitive to resolution in frequency and wavenumber space, and especially in the absence of tapering (see the following), poorer resolution rapidly leads to unusable data for narrow and strongly directional wave fields. Biases are most severe at low wavenumbers compared with that at the spectral peak.

Tapering the video cubes with a 3-D Hann window (e.g., [33]) lowers the biases to the velocity resolution level implied by the wave vector and frequency resolutions—$\delta c_{\delta k}$ and $\delta c_{\delta\omega}$ as defined in (4)—or even below. Indeed, the effect of ω resolution is mostly removed for tapered data (note that although biases are now subresolution, the resolution itself will eventually be too poor for purpose). This implies that in data acquisition, one should prioritize large areas rather than longer time series if spectral leakage is a problem.

The effect of spectral leakage is most pronounced in the k_x- and k_y-directions in the spectrum, and hence depends on the angle of propagation relative to these. The biases increase toward an angle of $\theta_0 = 45°$ for all wavenumbers outside a small range around the spectral peak $k = k_p$. This dependence on θ_0 is also mostly removed by tapering the data. Clearly, the observed current velocity cannot depend on which way the camera is held, meaning that comparison with results when the video is rotated, say, 45° could give a simple indication of the severity of spectral leakage problems.

For the case of a constant background current $\boldsymbol{U}$ with strengths up to $0.15\, c_0(k_p)$, a strong dependence on the angle between the current and the waves is observed. While the magnitudes of the spurious DSVs are mostly smaller than the implied velocity resolutions, we find a significant bias in the direction of the DSVs around $k = k_p$ when the waves propagate perpendicular to the current. This is also mitigated by tapering the data, but not removed in the case of a cross current.

A. Recommendations for Mitigation

Summarizing the outcome of our analysis from a practical viewpoint, we offer the following considerations to mitigate the errors and biases related to spectral leakage in remote sensing of currents from observed wave spectra. In the extraction of DSVs from the spectrum, the commonly used LS method is not recommended except if calculation cost is a severe restriction; a NSP procedure gives universally better results (other methods are also in use, but were not tested). Tapering the spatiotemporal data with a 3-D Hann window greatly reduces the mean and random biases and their dependence on spectral shape, spectral resolution, and camera orientation. When errors due to spectral leakage are suspected, rotating the camera (either the actual camera or the resulting images) by some intermediate angle before analysis (we used 30°) and comparing results could reveal whether long-range spectral leakage is causing spurious results, because leakage mainly occurs along the axes of the images. We find that the wavenumber resolution plays a more important role in the DSV biases than frequency resolution. The influence of the latter can be nearly eliminated by tapering the data. Increasing the spatial domain size to improve wavenumber resolution yields the largest improvement and should, therefore, be prioritized over longer time series if spectral leakage is a concern. The Doppler shift resolutions $\delta c_{\delta\omega}$ and $\delta c_{\delta k}$ defined in (4) are useful as conservative measures of errors and biases due to limited resolution and spectral leakage.

Appendix

A. Comparison of Window Functions

There exists a wide variety of window functions employed across disciplines, as discussed in, e.g., [33]. The suppression of the long-range spectral leakage is the most important beneficial effect of using a window function, but stronger

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suppression comes at the cost of increasing the short-range leakage. For example, the Blackman window has a central lobe that is about 20% wider than that of the Hann window, as can be seen in Fig. 1. To examine its detrimental effect, the extraction of DSVs has been repeated with a selection of commonly used window functions. As the results are very similar, we only show the results for the Hann window

$$w(t) = 0.5 - 0.5\cos(2\pi\, t/T), \tag{17}$$

the approximate Blackman window

$$w(t) = 0.42 - 0.5\cos(2\pi\, t/T) + 0.08\cos(4\pi\, t/T), \tag{18}$$

and the Kaiser–Bessel window

$$w(t) = \frac{1}{T} I_0\left(b\sqrt{1-(2\pi\, t/T)^2}\right) \tag{19}$$

with I_0 being the zeroth-order modified Bessel function of the first kind and b a free parameter controlling the width of the central lobe. In Fig. 10, the extracted DSVs in terms of their main and random bias are shown. The differences in the results are insignificant, except for the Blackman window showing systematically higher biases due to its wider central lobe.

B. Details of Numerical Implementation

Here follow further details on the implementation of the NSP method in Section III-C.

1) NSP, LS, and Discretized Data: Because of discretization, obtaining the spectral intensity P_i on a cylinder surface needs to be replaced with the spectral intensity in a volume around it, i.e., a cylinder shell, containing a wavenumber bin around k_i

$$F_i(\mathbf{k},\omega) = \begin{cases} \sqrt{P(\mathbf{k},\omega)}, & \text{if } |k-k_i| \le \tilde{\Delta}k \\ 0, & \text{otherwise.} \end{cases} \tag{20}$$

This also renders the characteristic function G_i [see (14)], a function of k. The integrals $\langle \ldots \rangle$ in (16) then imply an additional integration over k within the bin $k_i \pm \tilde{\Delta}k$.

The concept of the NSP method stays the same, with two details added: first, since the optimization parameter pair $\tilde{\boldsymbol{c}}$ is assumed constant within a wavenumber bin, the extracted DSV is a weighted average within that bin, effectively smoothing the function $\tilde{\boldsymbol{c}}(k)$. This also holds for the LS method, but does not change (21), as it only increases the number of triplets $(k_{x,j}, k_{y,j}, \omega_j)$. Second, a new free parameter is introduced with the bin width $2\tilde{\Delta}k$ that needs to be chosen carefully. In this work, we use $\tilde{\Delta}k = 2\delta k$, with $\delta k = 1/L$ the wavenumber resolution, which is a compromise between increased smoothing (too large $\tilde{\Delta}k$) and strong noise, occurring when too few pixels of the spectrum lie within a bin [one could employ an interpolation scheme to circumvent this; the algorithm used herein simply masks the data; see (20)].

Since the data are effectively averaged over k with a running average of width $2\tilde{\Delta}k$, the DSVs were extracted such that two consecutive shells have an overlap of 3/4, i.e., $k_{i+1} - k_i = \delta k$.

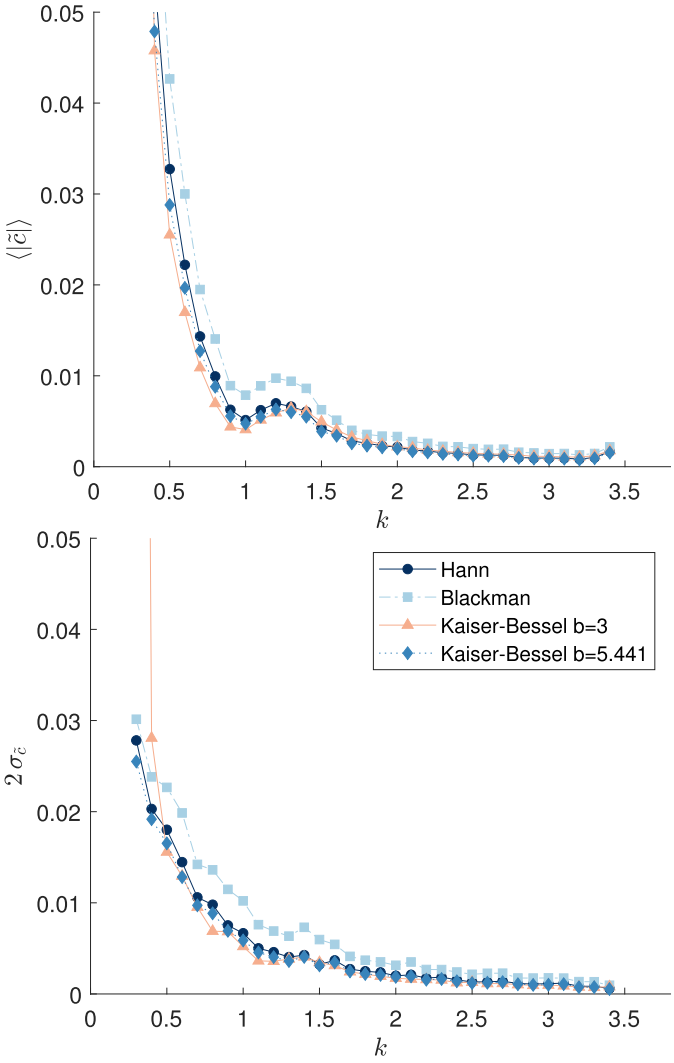


Fig. 10. DSVs in units of the peak phase velocity for a normal parameter combination ($\theta_0 = 90°$, $\Delta\theta = 60°$, $\gamma = 3.3$, $L = 10$, $T = 20$, and $U = 0$) obtained using the NSP method after tapering the data with different window functions. The top and bottom graphs show the DSVs $\tilde{\boldsymbol{c}}$ in terms of the mean (top) and standard deviation (bottom), respectively.

2) Width of the Characteristic Function: As mentioned in Section V, the characteristic function G (14) used in fitting the dispersion relation to the measured spectrum contains one free parameter, a, that determines its width. The choice of this parameter is somewhat delicate, as too small a value causes single intensity pixels (often outliers) or small high intensity regions to dominate the determination of the best fit. This is especially problematic for small ($k < 1$) wavenumbers, when the spectral leakage from a noise-enhanced pixel near the spectral peak at $k \approx 1$ causes a small high-intensity region on the cylinder shell, whereas the "real" spectral intensity, not originating from spectral leakage, is strongly broadened, thus having a larger total intensity but a smaller maximum intensity. In this case, the algorithm effectively ignores the real spectral intensity and leads to huge biases. On the other hand, too large a value for a also leads to an increased influence of spectral leakage, as a too wide Gaussian is insensitive to shifts in its position. A shift with no, or only small penalty to the overlap with the real spectral intensity, that increases the overlap with

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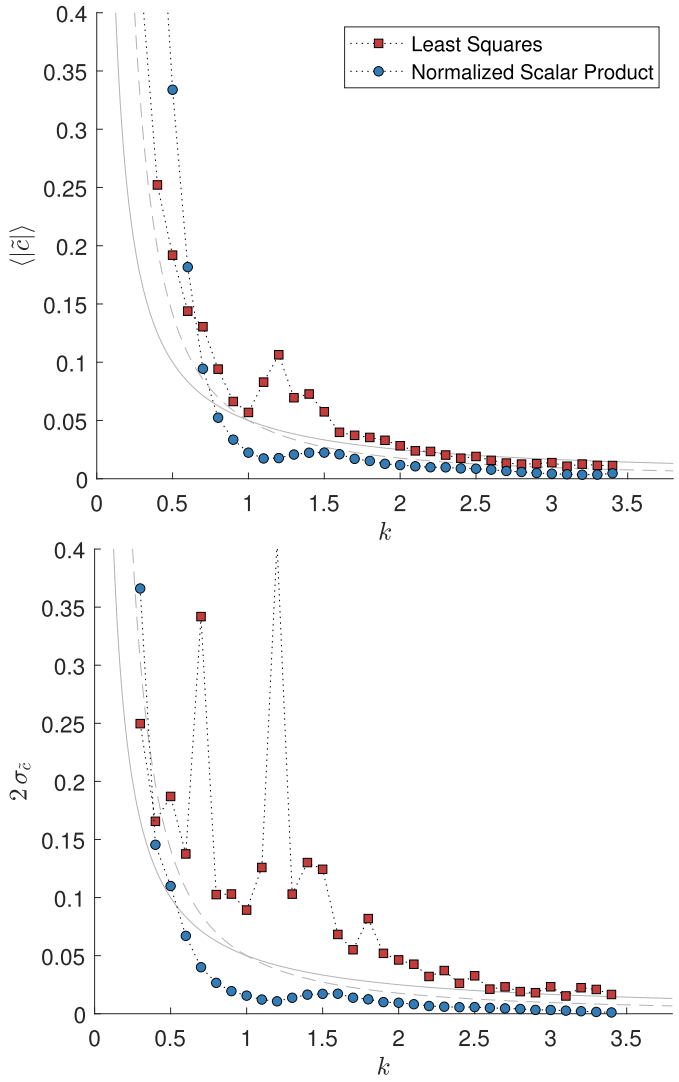


Fig. 11. DSVs in units of the peak phase velocity for the parameter set, case NSP–LS (see Table I), extracted using the NSP and an LS method. The top and bottom graphs show the DSVs $\tilde{\boldsymbol{c}}$ for quiescent water in terms of the mean (top) and standard deviation (bottom). The solid and dashed lines correspond to the velocity resolutions implied by frequency and wavenumber resolution, respectively.

intensity from leakage, is, therefore, more likely, also leading to biases.

To decide on a good compromise, we fit the characteristic function to the Fourier transform of a Hann window, with a as a free parameter. The result is $a \approx 1.62\,\delta\omega$. To ensure the width is not too small, this value is roughly doubled to $a = 4\,\delta\omega$ (values for $a/\delta\omega$ between 2 and 6 were tried as well, but the results were most stable between 3 and 5).

Note that if spectral leakage occurred only in the ω-direction, $a = \delta\omega$ would be the ideal choice. However, as this is not the case, broadening in the ω-direction originates not only from leakage in the ω-direction, but also from leakage in the k-direction. The steeper the dispersion relation $\omega_{\mathrm{DR}}(k)$, the more this effective ω leakage from k leakage increases. A possible improvement of the used NSP algorithm would, therefore, be to use an adaptive a, that increases with $|\partial\omega/\partial k|$. In the interest of limiting the parameter space, it was deemed necessary to stick to a single value for a.

C. Comparison of NSP and a Basic LS Algorithm

Simple LS-based methods have, by now, been discarded in most applications. However, their low-computational cost and ease of implementation are benefits to consider. We, therefore, repeated the DSV extraction using LS and compared the results and performance. The LS method used, such as the NSP method described in Section III-C, first singles out the spectral intensity on a cylinder surface with radius k. We discard data points with an intensity $P < 0.2$ (after normalization) and obtain a list of $(k_{x,j}, k_{y,j}, \omega_j)$. On these, the cost function

$$C(\tilde{\boldsymbol{c}}) = \sum_j \left(\omega_{\mathrm{DR}}(\boldsymbol{k}_j; \tilde{\boldsymbol{c}}) - \omega_j\right)^2 \tag{21}$$

is minimized to obtain the DSV $\tilde{\boldsymbol{c}}$. Repeating this for a list of wavenumbers k yields the desired value of $\tilde{\boldsymbol{c}}$ for each value of k. As in the NSP method, the minimization/optimization step is performed using the Nelder–Mead simplex method [35].

We find that the LS method runs significantly faster (up to a factor of 3); clearly, the difference in cost will depend on the hardware used as well as the implementation. In terms of accuracy and precision, we find that the LS method performed consistently worse, and never better, than the NSP method. Fig. 11 shows an example result for a test case in quiescent water (see Table I, case NSP–LS, for all parameters). As can be seen, the DSVs obtained via the LS method can show both mean and random biases that exceed the implied velocity resolutions $\delta c_{\delta k}$ and $\delta c_{\delta\omega}$ for most wavenumbers, while the NSP method delivers subresolution DSVs for all $k > 1$. It is worth pointing out that for wavenumbers $k < 1$, the NSP performs worse than the LS method. However, the random bias in that range is so large as to make both methods unusable. Note that a more advanced LS method may perform with similar accuracy and precision, as is indicated in, e.g., [10]. This would, presumably, lead to a computational cost similar to that of the NSP method, eliminating the advantage. An iterative LS method has been compared with NSP by Huang and Gill [10]. For waves on currents without vertical shear, the methods performed similarly.

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